

AEO «Nazarbayev intellectual schools»
Center of pedagogical measurements



EXAMPLE CANDIDATE RESPONSE OF EXTERNAL SUMMATIVE ASSESSMENT

12 grade

CHEMISTRY
Paper 3

Astana, 2024 year

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QUESTION 1(A)

1 A student determines the percentage by mass of iron(II) ions in an iron(II) salt by titration with acidified potassium manganate(VII), KMnO_4 , using the following steps.

Step 1 Accurately weighs 4.50 g of the iron(II) salt and pour it into a 250 cm^3 beaker.

Step 2 Adds distilled water to the beaker until all the unknown iron(II) salt dissolves.

Step 3 Transfers this solution into the 250 cm^3 graduated flask, fills it with water, and labels it as **B**.

Step 4 Transfers 25.0 cm^3 of solution **B** into a conical flask.

Step 5 Adds 10 cm^3 of 1.00 mol dm^{-3} sulfuric acid to the conical flask.

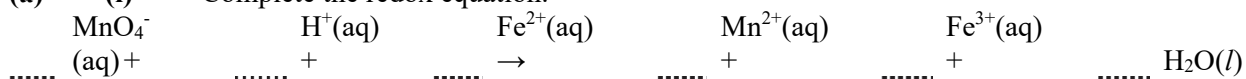
Step 6 Fills a burette with the 0.0200 mol dm^{-3} solution $\text{KMnO}_4(\text{aq})$.

Step 7 Records the initial burette reading.

Step 8 Titrates solution **B** against the $\text{KMnO}_4(\text{aq})$ until the solution changes to a permanent pink colour.

Step 9 Records the final burette reading.

(a) (i) Complete the redox equation.



[1]

(ii) Use changes in oxidation states to identify the reducing agent and the oxidising agent in this reaction.

reducing agent

oxidising agent

[2]

Mark Scheme

Question	Answer	Mark	Additional Guidance
1(a) (i)	$\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Mn}^{2+} (\text{aq}) + 5\text{Fe}^{3+} (\text{aq}) + 4\text{H}_2\text{O} (\text{l})$	[1]	
1(a) (ii)	Reducing agent – Fe^{2+} Oxidising agent – MnO_4^-	[2]	

Example candidate response—grade A

1.

(a) (i) Complete the redox equation.

$$\text{MnO}_4^-(\text{aq}) + \dots \text{H}^+(\text{aq}) + \dots \text{Fe}^{2+}(\text{aq}) \rightarrow \dots \text{Mn}^{2+}(\text{aq}) + \dots \text{Fe}^{3+}(\text{aq}) + \dots \text{H}_2\text{O}(\text{l})$$

[1] ✓ 1

(ii) Use changes in oxidation states to identify the reducing agent and the oxidising agent in this reaction.

reducing agent $\text{Fe}^{2+}(\text{aq})$

oxidising agent $\text{MnO}_4^-(\text{aq})$

[2] ✓ 2

Examiner's comment

The candidate earned three out of three marks for this question. He correctly determined the coefficients in the oxidation-reduction reaction and identified both the reducing and oxidising agents in the equation.

Example candidate response—grade C

1.

(a) (i) Complete the redox equation.

$$\text{MnO}_4^-(\text{aq}) + \dots \text{H}^+(\text{aq}) + \dots \text{Fe}^{2+}(\text{aq}) \rightarrow \dots \text{Mn}^{2+}(\text{aq}) + \dots \text{Fe}^{3+}(\text{aq}) + \dots \text{H}_2\text{O}(\text{l})$$

[1] ✗

(ii) Use changes in oxidation states to identify the reducing agent and the oxidising agent in this reaction.

reducing agent Fe^{+2} ✓ to Fe^{+3} (iron salt)

oxidising agent MnO_4^- ✓ (Mn^{+7} to Mn^{+2})

[2] ✓ 2

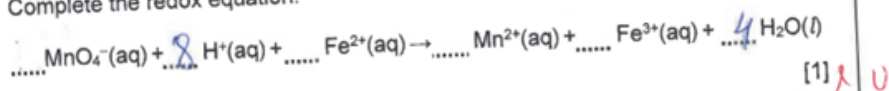
Examiner's comment

The candidate earned two out of three marks for this question. He did not balance the redox equation, resulting in a loss of one mark. The candidate was able to identify both the reducing and oxidising agents in the equation, for which he was awarded two marks.

Example candidate response—grade E

1.

(a) (i) Complete the redox equation.



[1] X U

(ii) Use changes in oxidation states to identify the reducing agent and the oxidising agent in this reaction.

reducing agent Fe

oxidising agent Mn

[2] X 0

Examiner's comment

The candidate did not earn any marks because he failed to determine the correct coefficients in the oxidation-reduction equation and did not indicate the oxidation states of the reducing and oxidising agents.

QUESTION 1(B)(I)

(b) The student obtains the results shown in the table.

Complete the table and state which titrations you use to calculate the mean titre.

Calculate the mean titre without including the rough titration.

titration number	1	2	3	4
initial (burette) reading/ cm ³	0.0	1.0	1.0	0.5
final (burette) reading/ cm ³	27.8	28.0	27.9	27.9
volume of solution added/ cm ³				

titrations used

mean titre = cm³
[3]

Mark Scheme

1(b) (i)	for completing table: 27.8(0); 27.0(0); 26.9(0); 27.4(0) titrations used – 2 and 3 mean titre – 27.00 / 26.95	1 1 1 [3]	accept mean quoted to 1, 2 or 3 decimal places correctly rounded
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Example candidate response—grade A

(b) The student obtains the results shown in the table.

(i) Complete the table and state which titrations you use to calculate the mean titre.

Calculate the mean titre without including the rough titration.

titration number	1	2	3	4
initial (burette) reading/ cm ³	0.0	1.0	1.0	0.5
final (burette) reading/ cm ³	27.8	28.0	27.9	27.9
volume of solution added/ cm ³	27.8	27.0	26.9	27.4

titrations used 2 and 3

mean titre = $\frac{27.0 + 26.9}{2} = 26.95$ cm³
[3]

Examiner's comment

The candidate earned three out of three marks for this question. He correctly calculated the volume of solution used in four experiments and was awarded one mark. He also earned two marks for correct selection of titres with a difference of ± 0.1 cm³ and calculation of the mean titre.

Example candidate response—grade C

(b) The student obtains the results shown in the table.

(i) Complete the table and state which titrations you use to calculate the mean titre.

Calculate the mean titre without including the rough titration.

titration number	1	2	3	4
initial (burette) reading/cm ³	0.0	1.0	1.0	0.5
final (burette) reading/cm ³	27.8	28.0	27.9	27.9
volume of solution added/cm ³	27.8	27.0	26.9	27.4

titrations used 2, 3, 4 X

mean titre = $\frac{27.0 + 26.9 + 27.4}{3} = 27.1$ cm³ ecf ✓ [3]

Examiner's comment

The candidate earned two out of three marks for this question. He correctly calculated the volume of solution used in four experiments, earning one mark, but failed to determine the correct titres with a difference of ± 0.1 , resulting in a loss of one mark. However, he correctly calculated the mean titre, so ecf was applied to his answer, and he was awarded one mark.

Example candidate response—grade E

(b) The student obtains the results shown in the table.

(i) Complete the table and state which titrations you use to calculate the mean titre.

Calculate the mean titre without including the rough titration.

titration number	1	2	3	4
initial (burette) reading/cm ³	0.0	1.0	1.0	0.5
final (burette) reading/cm ³	27.8	28.0	27.9	27.9
volume of solution added/cm ³				

titrations used ~~2, 3, 4~~ 1, 3, 4 X

mean titre = $\frac{27.8 + 27.9 + 27.9}{3} = \frac{83.6}{3} = 27.9$ cm³ ecf ✓ [3]

Examiner's comment

The candidate earned one out of three marks for this question. He failed to calculate the volume of titrant used in four experiments and incorrectly determined the titres. However, he calculated the mean titre based on the volumes he selected, earning one mark according to ECF.

QUESTION 1(B)(II–V)

- (ii) Use the titration results to calculate the number of moles of MnO_4^- .

..... mol [1]

- (iii) Calculate the number of moles of iron(II) salt in 250 cm^3 of solution.

..... mol [2]

- (iv) Assume that in 1 mole of the iron(II) salt contains 1 mole of iron(II) ions. Calculate the mass of Fe^{2+} ions in iron (II) salt.

..... g [1]

- (v) Calculate the percentage of mass of Fe^{2+} ions from the total mass of iron(II) salt.

..... % [1]

Mark Scheme

1(b) (ii)	Moles of $\text{MnO}_4^- = 0.0200 \times \frac{27}{1000} = 5.4 \times 10^{-4}$ / 5.39×10^{-4} / 0.000539	[1]	ecf from 1(b)(i) and 1(a)(i)
1(b) (iii)	Moles in $25 \text{ cm}^3 = 5.4 \times 10^{-4} \times 5 = 2.7 \times 10^{-3}$ / 0.0027/ 2.695×10^{-3} / 0.002695 Moles in $250 \text{ cm}^3 = 2.7 \times 10^{-2}$ / $0.027 / 2.695 \times 10^{-2}$ / 0.02695	[2]	Accept correct answer award 2 marks ecf from 1(b)(ii)
1(b) (iv)	(mass of Fe^{2+}) = (moles of Fe^{2+}) $\times$ 56 (55.8) = $0.027 \times 56 = 1.51 \text{ g}$ / 1.50 g	[1]	ecf from 1(b)(iii)
1(b) (v)	percentage by mass = $\frac{1.51}{4.50} \times 100\% = 33.6\%$ / 33.418%	[1]	Accept 33.55%/33.33% ecf from 1(b)(iv)

Example candidate response—grade A

	(ii) Use the titration results to calculate the number of moles of MnO_4^- .	For Examiner's Use
	$c(\text{MnO}_4^-) = 0,02 \frac{\text{mol}}{\text{dm}^3}$	
mean titre $\Rightarrow$	$V(\text{MnO}_4^-) = 26,95 \text{ cm}^3 = 26,95 \times 10^{-3} \text{ dm}^3$	
	$n(\text{MnO}_4^-) = 26,95 \times 10^{-3} \text{ dm}^3 \times 0,02 \frac{\text{mol}}{\text{dm}^3} = 5,39 \times 10^{-4} \text{ mol}$	1
	 $5,39 \times 10^{-4}$ mol [1]	
	(iii) Calculate the number of moles of iron(II) salt in 250 cm^3 of solution.	
ratio $1:5$ $\Rightarrow$	$n(\text{Fe}^{2+}) = 5,39 \times 10^{-4} \text{ mol} \times 5 = 2,695 \times 10^{-3} \text{ mol (in } 25 \text{ cm}^3)$	
	$n(\text{Fe}^{2+}) = 2,695 \times 10^{-3} \text{ mol} \times 10 = 0,02695 \text{ mol (in } 250 \text{ cm}^3)$	
	 $0,02695$ mol [2]	2
	(iv) Assume that in 1 mole of the iron(II) salt contains 1 mole of iron(II) ions. Calculate the mass of Fe^{2+} ions in iron(II) salt.	
	$1 \text{ mole of Fe}^{2+} \text{ salt} \rightarrow 1 \text{ mole of Fe}^{2+} \text{ ions}$	
$m = n \cdot M$	$m(\text{Fe}^{2+}) = 55,8 \frac{\text{g}}{\text{mol}} \times 0,02695 \text{ mol} = 1,50381 \text{ g} \approx 1,50 \text{ g}$	
$M = 55,8 \text{ g/mol}$	 g [1]	1
	(v) Calculate the percentage of mass of Fe^{2+} ions from the total mass of iron(II) salt.	
	$\frac{4,50 \text{ g (iron(II) salt)}}{1,50 \text{ g (Fe}^{2+} \text{ ions)}} - 100\%$	
	$x = \frac{1,50 \text{ g} \times 100\%}{4,50 \text{ g}} \approx 33,3\%$	
	 $33,3$ % [1]	1

Examiner's comment

The candidate earned five out of five marks for this question. He correctly calculated the amount of MnO_4^- . Considering the coefficients in the reaction equation from question (a)(i), he correctly calculated the amount of iron(II) ions by proportion and converted this value to a volume of 250 cm^3 . He also correctly determined the mass of iron(II) ions and calculated its percentage content in the salt.

Example candidate response—grade C

(ii) Use the titration results to calculate the number of moles of MnO_4^-. $V = 27,1 \text{ cm}^3$ $C = 0,02$ $n = C \cdot V = 27,1 \cdot 10^{-3} \cdot 0,02 = 5,42 \cdot 10^{-4}$ $5,42 \cdot 10^{-4}$ mol [1]	For Examiner's Use 1
(iii) Calculate the number of moles of iron(II) salt in 250 cm^3 of solution. $V = 250 \text{ cm}^3$ $n = 250 \cdot 10^{-3}$ $n = C \cdot V$ $C = 0,02$ $n = 0,005$ $m = 4,50 \text{ g}$ $n = \frac{m}{M}$ $\frac{4,5}{56} = 0,08$ $0,08$ mol [2]	0
(iv) Assume that in 1 mole of the iron(II) salt contains 1 mole of iron(II) ions. Calculate the mass of Fe^{2+} ions in iron(II) salt. $m = n \cdot M$ $m = 0,005 \cdot 56 = 0,28$ $0,28$ g [1]	1
(v) Calculate the percentage of mass of Fe^{2+} ions from the total mass of iron(II) salt. $\frac{0,28}{4,5} \cdot 100\% = 6,22$ $6,22\%$ [1]	1

Examiner's comment

The candidate earned three out of five marks for this question. He correctly calculated the amount of MnO_4^- , earning one mark. However, when calculating the amount of iron(II) ions, he did not account for the coefficients in the reaction equation and made an error in converting this value to a volume of 250 cm^3 , resulting in a loss of two marks. Despite this, using the obtained value, the candidate was then able to correctly calculate the mass and percentage of iron(II) ions in the salt, allowing for the application of ecf, and was awarded two marks.

Example candidate response—grade E

(ii) Use the titration results to calculate the number of moles of MnO_4^- .

$$C(\text{KMnO}_4) = 0,0200 \text{ mol/dm}^3 \quad C = \frac{n}{V} \Rightarrow n = C \cdot V$$

$$V = 2,9 \text{ cm}^3 \quad C \cdot \frac{n}{V} \Rightarrow n = 0,0200 \cdot 2,9 \cdot 10^{-3} = 5,8 \cdot 10^{-5}$$

..... mol [1]

(iii) Calculate the number of moles of iron(II) salt in 250 cm^3 of solution.

$$m(\text{iron(II) salt}) = 4,50 \text{ g} \quad n = 0,5 \text{ moles} \quad \frac{0,25}{4,50}$$

$$V(250 \text{ cm}^3) = 0,25 \text{ dm}^3 \quad 10^{-2}$$

$$0,5 \cdot 10^{-2} \text{ mol [2]}$$

(iv) Assume that in 1 mole of the iron(II) salt contains 1 mole of iron(II) ions.
Calculate the mass of Fe^{2+} ions in iron(II) salt.

$$M_r(842) \quad 0,5 \cdot 10^{-2} \cdot 2 = 0,01$$

$$1:1$$

..... g [1]

(v) Calculate the percentage of mass of Fe^{2+} ions from the total mass of iron(II) salt.

$$\frac{4,50 - 100\%}{93 - ?} \quad x = \frac{0,5 \cdot 100}{4,5}$$

..... % [1]

Examiner's comment

The candidate earned one out of five marks for this question. The amount of MnO_4^- was calculated incorrectly. The candidate did not account for the coefficients in the reaction equation when calculating the amount of iron(II) ions and made an error in determining the number of moles in 250 cm^3 of solution. The candidate also failed to correctly calculate the mass of iron(II) ions in the salt. However, he was able to use the obtained value to calculate the percentage of the iron(II) ions in the salt, allowing for the application of ecf, and was awarded one mark.

QUESTION 1(C)

(c) The student decides that the change in colour during the titration is the equivalence point.

(i) State and explain whether the student is correct.

.....
 [1]

(ii) The error in a single burette reading is $\pm 0.05 \text{ cm}^3$.

What is the percentage error in the titre volume calculated in (b)(i)? Give your answer to 2 significant figures.

percentage error = % [1]

The student uses a measuring cylinder to transfer solution **B** to the conical flask.

Suggest a more accurate piece of apparatus to use instead of a measuring cylinder.

..... [1]

Mark Scheme

Question	Answer	Mark	Additional Guidance
1(c)(i)	Student was incorrect. Equivalent point occurs before end point, colour change occurs at the end point.	[1]	owtte
1(c)(ii)	percentage error = $\frac{0.1}{27.00} \times 100\% = 0.37\%$	[1]	Accept 26,95 instead of 27 in calculation ecf from 1(b)(i)
1(c)(iii)	pipette	[1]	Accept burette/ electronic dropper

Example candidate response—grade A

(c) The student decides that the change in colour during the titration is the equivalence point.

(i) State and explain whether the student is correct.

No, it is a end point - when the color of solution changes. Equivalence point is when number of molecules of oxidizing and reducing agent is equal [1]

(ii) The error in a single burette reading is $\pm 0.05 \text{ cm}^3$. What is the percentage error in the titre volume calculated in (b)(i)? Give your answer to 2 significant figures.

$$\frac{0.05 + 0.05}{26.95} \times 100 = 0.37$$

percentage error = 0.37 % [1]

(iii) The student uses a measuring cylinder to transfer solution B to the conical flask. Suggest a more accurate piece of apparatus to use instead of a measuring cylinder.

bulb pipette [1]

Examiner's comment

The candidate earned three out of three marks for this question. He made a correct judgment on the given statement and clearly explained the difference between the end and equivalence points. He also correctly calculated the percentage error in the burette reading based on the average titre and suggested a more accurate equipment.

Example candidate response—grade C

(c) The student decides that the change in colour during the titration is the equivalence point.

(i) State and explain whether the student is correct.

the colour changes, but not at equivalence point, it changes at endpoint. equivalence point means that the number of moles of reducing and oxidizing agent becomes equal. After it at endpoint the color changes to show the color, neutralization is occurs [1]

(ii) The error in a single burette reading is $\pm 0.05 \text{ cm}^3$. What is the percentage error in the titre volume calculated in (b)(i)? Give your answer to 2 significant figures.

$$\frac{0.05}{26.95} \times 100 = 0.185 \approx 0.19$$

percentage error = 0.19 % [1]

(iii) The student uses a measuring cylinder to transfer solution B to the conical flask. Suggest a more accurate piece of apparatus to use instead of a measuring cylinder.

using graduated pipette or usual pipette, which has the error reading ± 0.04 . [1]

[Total: 14]

Examiner's comment

The candidate earned two out of three marks for this question. He made a correct judgement on the judged on the given statement and explained the difference between the end and equivalence points. He suggested a more accurate equipment to measure the volume. The candidate incorrectly calculated the percentage error in the burette reading by failing to multiply ± 0.05 by 2, resulting in a loss of one mark.

Example candidate response—grade E

- (c) The student decides that the change in colour during the titration is the equivalence point.

- (i) State and explain whether the student is correct.

Student is correct in titration experiment colour change is the equivalence point. [1]

- (ii) The error in a single burette reading is $\pm 0.05 \text{ cm}^3$. What is the percentage error in the titre volume calculated in (b)(i)? Give your answer to 2 significant figures.

$$\frac{0.0200}{0.05} \times 100\%$$

percentage error = 40 % [1]

- (iii) The student uses a measuring cylinder to transfer solution B to the conical flask. Suggest a more accurate piece of apparatus to use instead of a measuring cylinder.

To take data student should check the measurement of cylinder at human eyes point, calculate correctly to take correct results. [Total: 14]

Examiner's comment

The candidate did not earn any marks for this question. He made an incorrect judgment because he failed to correctly explain the equivalence point, incorrectly calculated the percentage error, resulting in a loss of two marks. He also failed to suggest an apparatus to obtain more accurate results.

QUESTION 2(A),(B)(I)

- 2 The enthalpy change, ΔH_r , for the decomposition of calcium carbonate to calcium oxide cannot be measured directly.



The student determines the enthalpy change for the reactions of calcium carbonate and calcium oxide with hydrochloric acid. Excess hydrochloric acid will be used in both experiments.

Then the student uses Hess' Law to calculate the enthalpy change for the reaction above.

State Hess' Law.

.....
..... [1]

The student carries out two separate experiments and uses the results of these experiments to determine the enthalpy change of reaction for the decomposition of CaCO_3 by following these instructions.

Experiment 1 with $\text{CaCO}_3(\text{s})$ and $\text{HCl}(\text{aq})$

Step 1 Uses a measuring cylinder to place 50 cm^3 of 3.0 mol dm^{-3} hydrochloric acid into a 250 cm^3 glass beaker.

Step 2 Records the temperature of the hydrochloric acid.

Step 3 Places 2.5 g of calcium carbonate into the hydrochloric acid.

Step 4 Stirs the mixture using a thermometer and records the highest temperature obtained.

Step 5 Records the results in a table.

Student obtains the following results from Experiment 1.

Complete the table by calculating the temperature change.

Experiment	$\text{CaCO}_3(\text{s})$
temperature of acid at start / $^{\circ}\text{C}$	24.5
highest temperature of reaction mixture / $^{\circ}\text{C}$	28.0
temperature change / $^{\circ}\text{C}$	

[1]

Mark Scheme

2(a)	The enthalpy (energy) change for converting reactants into products is the same regardless of the route taken	[1]	Idea of the total enthalpy change of a reaction is independent of the reaction route
2 (b) (i)	3.5°C	[1]	

Example candidate response—grade A

- 2 The enthalpy change, ΔH_r , for the decomposition of calcium carbonate to calcium oxide cannot be measured directly.



The student determines the enthalpy change for the reactions of calcium carbonate and calcium oxide with hydrochloric acid. Excess hydrochloric acid will be used in both experiments.

Then the student uses Hess' Law to calculate the enthalpy change for the reaction above.

- (a) State Hess' Law.

enthalpy change of reaction is independent
on route taken.

[1]

- (b) The student carries out two separate experiments and uses the results of these experiments to determine the enthalpy change of reaction for the decomposition of CaCO_3 by following these instructions.

- (i) Complete the table by calculating the temperature change.

Experiment	$\text{CaCO}_3(\text{s})$
temperature of acid at start / °C	24.5
highest temperature of reaction mixture / °C	28.0
temperature change / °C	3.5

[1]

Examiner's comment

The candidate earned two out of two marks for this question. He provided the correct definition of Hess' Law and correctly calculated the temperature change.

Example candidate response—grade C

- 2 The enthalpy change, ΔH_r , for the decomposition of calcium carbonate to calcium oxide cannot be measured directly.



The student determines the enthalpy change for the reactions of calcium carbonate and calcium oxide with hydrochloric acid. Excess hydrochloric acid will be used in both experiments.

Then the student uses Hess' Law to calculate the enthalpy change for the reaction above.

- (a) State Hess' Law.

enthalpy change can be measured
by summing up the enthalpy change of the steps

[1]

- (b) The student carries out two separate experiments and uses the results of these experiments to determine the enthalpy change of reaction for the decomposition of CaCO_3 by following these instructions.

- (i) Complete the table by calculating the temperature change.

Experiment	$\text{CaCO}_3(\text{s})$
temperature of acid at start / °C	24.5
highest temperature of reaction mixture / °C	28.0
temperature change / °C	3.5

[1]

Examiner's comment

The candidate earned one out of two marks for this question. He failed to correctly define Hess' Law, resulting in a loss of one mark. However, he correctly calculated the temperature change.

Example candidate response—grade E

- 2 The enthalpy change, ΔH_r , for the decomposition of calcium carbonate to calcium oxide cannot be measured directly.



The student determines the enthalpy change for the reactions of calcium carbonate and calcium oxide with hydrochloric acid. Excess hydrochloric acid will be used in both experiments.

Then the student uses Hess' Law to calculate the enthalpy change for the reaction above.

- (a) State Hess' Law.

overall change of heat in cycle of reactions of CaCO_3 decom-
position X [1]

- (b) The student carries out two separate experiments and uses the results of these experiments to determine the enthalpy change of reaction for the decomposition of CaCO_3 by following these instructions.

- (i) Complete the table by calculating the temperature change.

Experiment	$\text{CaCO}_3(\text{s})$
temperature of acid at start / °C	24.5
highest temperature of reaction mixture / °C	28.0
temperature change / °C	-3.5 X

$$24.5 - 28.0 \rightarrow -3.5$$

[1]

Examiner's comment

The candidate did not earn any marks for this question. He did not correctly define Hess' Law and also made an error in calculating the temperature change, failing to correctly apply the expression $\Delta T = T_2 - T_1$.

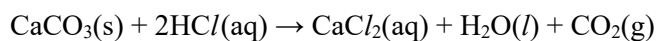
QUESTION 2(B)(II),(III)

2b (ii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
Assume the reaction mixtures has a density of 1.0 g cm^3 .

Use the temperature change in (b)(i) to calculate the heat released in Experiment 1.

heat released = J [1]

(iii) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_1 .



enthalpy change = kJ mol^{-1} [2]

Mark Scheme

2 (b) (ii)	735J	[1]	accept 0.294J Ignore Sign
2 (b) (iii)	0.025 mole - 29.4 kJ mol^{-1}	[2]	Do not accept 29.4 ecf from 2(b)(ii)

Example candidate response—grade A

(ii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
Assume the reaction mixtures has a density of 1.0 g cm^3 .

Use the temperature change in (b)(i) to calculate the heat released in Experiment 1.

$$q = mc\Delta t = 50 \cdot 4.2 \cdot 3.5 = 735 \text{ J}$$

$$m(\text{HCl}) = 50 \text{ cm}^3 = 50 \text{ g}$$

heat released = 735 J [1]

For
Examiner's
Use

(iii) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_1 .



$\Delta H = \frac{Q}{n} = \frac{0,735}{0,025} = 29,4$ reaction is exothermic, so $\Delta H = -29,4 \text{ kJ mol}^{-1}$

$n = \frac{2,5}{100,1} = 0,025$ ✓
 $n = 0,024975$

enthalpy change = -29,4 kJ mol^{-1} [2] ✓

Examiner's comment

The candidate earned three out of three marks for this question. He successfully completed all calculations, and correctly determined both the amount of heat released and the enthalpy change for the reaction in kJ/mol .

Example candidate response—grade C

- (ii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
 Assume the reaction mixture has a density of 1.0 g cm^{-3} .

For
Examiner's
Use

Use the temperature change in (b)(i) to calculate the heat released in Experiment 1.

$Q = m \cdot c \cdot \Delta t = 50 \text{ g} \cdot 4,2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1} \cdot 3,5 ^\circ\text{C} = 735 \text{ J}$

$m = \rho \cdot V = 1.0 \text{ g cm}^{-3} \cdot 50 \text{ cm}^3 = 50 \text{ g}$

heat released = 735 J [1] ✓

- (iii) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_1 .



$n(\text{CaCO}_3) = \frac{2,5 \text{ g}}{100 \text{ g mol}^{-1}} = 0,025 \text{ mol}$ ✓ $\Delta H = \frac{Q}{n} = \frac{0,735 \text{ kJ}}{0,025 \text{ mol}} = 29,4$

enthalpy change = 29,4 kJ mol^{-1} [2] ✗

Examiner's comment

The candidate earned two out of three marks for this question. He correctly determined the amount of heat released. However, when calculating the enthalpy change for the reaction, he made an error with the sign of the enthalpy value, resulting in a loss of one mark despite the accurate calculation.

Example candidate response—grade E

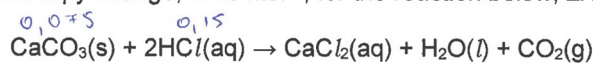
- (ii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
Assume the reaction mixture has a density of 1.0 g cm^3 .

Use the temperature change in (b)(i) to calculate the heat released in Experiment 1.

$$Q = m \cdot c \cdot \Delta T = 50 \cdot 4.2 \cdot 3.5 = 735 \text{ J}$$

heat released = 735 J [1]

- (iii) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_1 .



$$\Delta H = \frac{Q}{n} = \frac{\frac{735}{1000}}{0,15} = +4,9$$

$$n = \frac{50}{1000} \cdot 3.0 = 0,15 \text{ mol}$$

enthalpy change = +4,9 kJ mol^{-1} [2]

Examiner's comment

The candidate earned one out of three marks for this question. He correctly determined the amount of heat released during the reaction. However, when calculating the enthalpy change for the reaction, he calculated the substance in excess and indicated the incorrect sign for the enthalpy, leading to a loss of marks.

QUESTION 2(C)(I),(II)

(c) Experiment 2 with CaO(s) and HCl(aq)

The student:

Step 1 Uses a measuring cylinder to place 50 cm³ of 3.0 mol dm⁻³ hydrochloric acid into a 250 cm³ glass beaker.

Step 2 Records the temperature of the hydrochloric acid every minute for a total of eight minutes.

Step 3 Records the results in the table.

Step 4 At exactly 2.5 minutes adds 1.4 g of the calcium oxide into the hydrochloric acid.

time/minutes	0	1	2		3	4	5	6	7	8
temperature of mixture/ °C	24.0	24.5	24.5		35.0	37.5	36.0	34.5	35.0	31.5

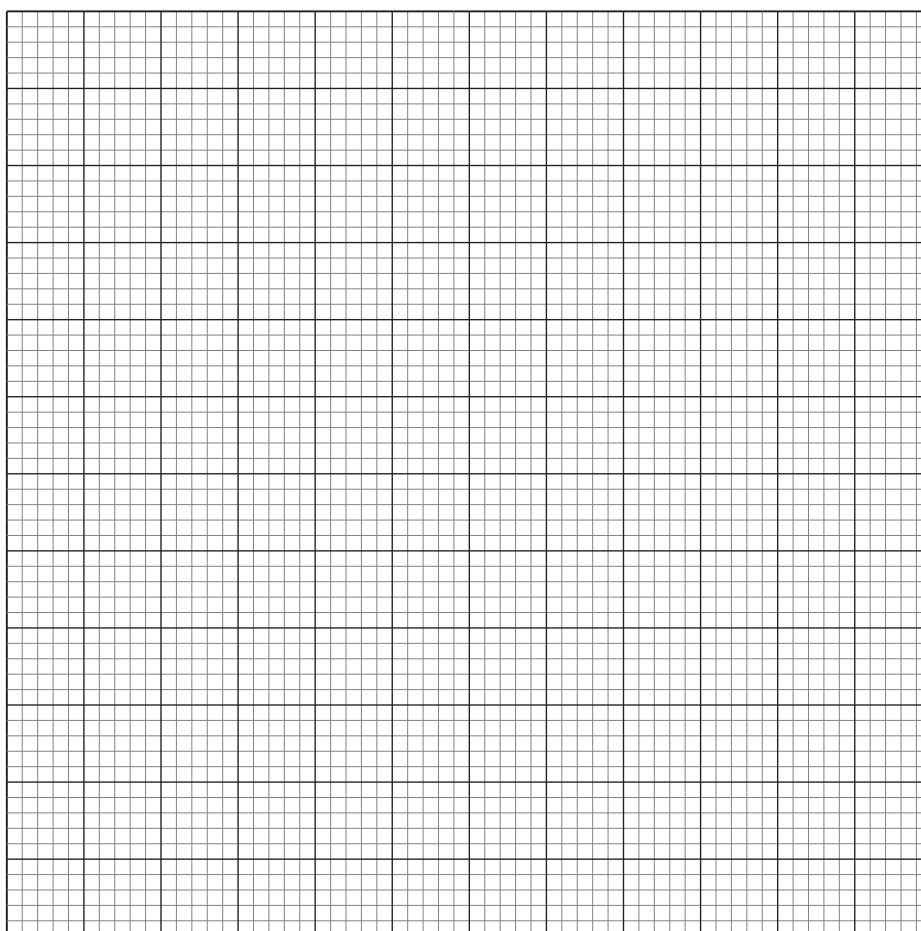
Use the results from Experiment 2 and plot a graph of temperature against time on the grid below.

Draw two lines of best fit for Experiment 2 through the points on your graph.

The first line is for the first three temperatures during the first two minutes.

The second line is for the last six temperatures during the last five minutes.

Extrapolate the lines to estimate the temperatures at 2.5 minutes and the temperature change, ΔT , at 2.5 minutes.



Calculate the maximum temperature change at 2.5 minutes.

$\Delta T =$ °C [4]

Circle the most anomalous point on your graph and suggest a reason for the anomalous result.

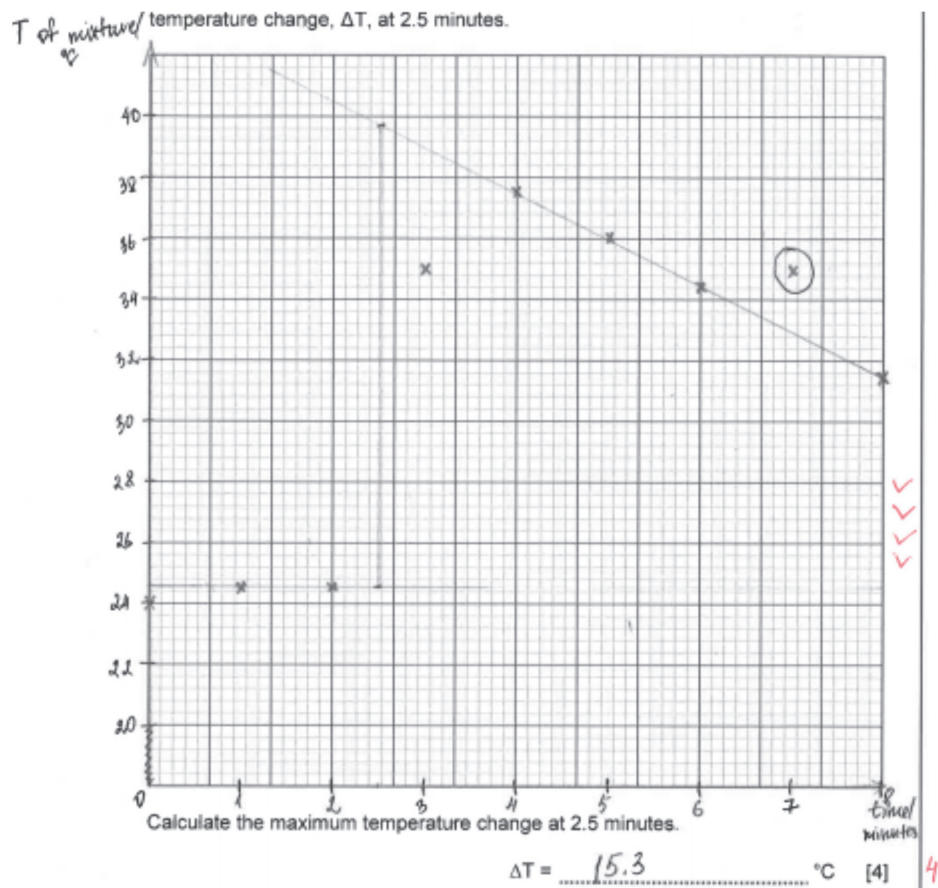
.....

..... [1]

Mark Scheme

2 (c) (i)	correct labels and suitable linear scales for both axes all points plotted to within half a square appropriate lines of best fit drawn. AND either a straight line / smooth curve after the max T Lines extrapolated and correct value of $\Delta T = 15,5 \pm 2$	1 1 1 1 [max 4]	ecf from straight lines on the plotted points
2 (c) (ii)	Anomalous points are marked. Not consequently stirring the mixture Or Heat loss (for anomalous point at 3rd minute) Or Change the conditions of experiment	[1]	Owtte Do not accept Human error

Example candidate response—grade A



- (ii) Circle the most anomalous point on your graph and suggest a reason for the anomalous result.

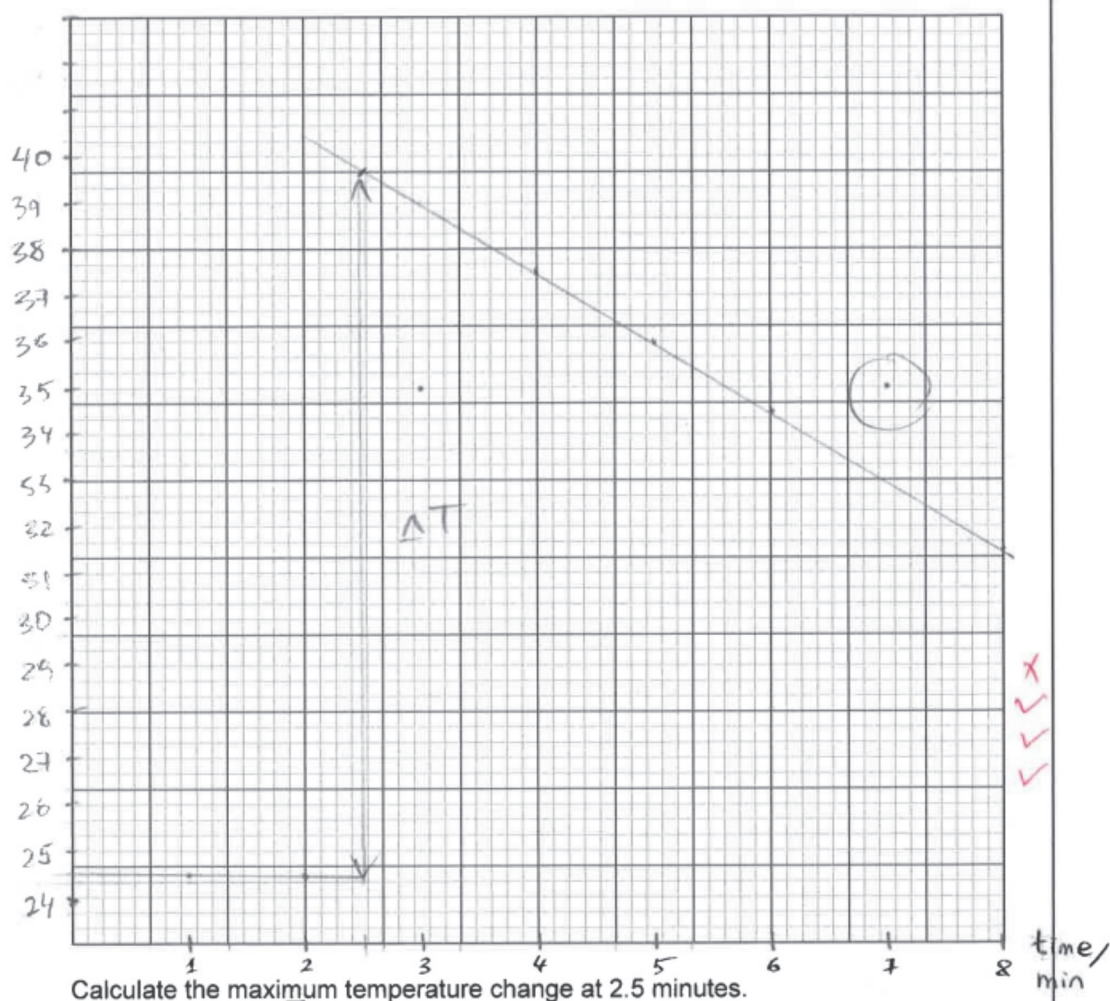
some amount of ~~the~~ catalyst can be left on the walls of the beaker and at 7 minutes they dropped into the solution to give a higher temperature which is anomalous.

Examiner's comment

The candidate earned five out of five marks for this question. He labelled the X and Y axes with the correct units of measurement and scale (earning one mark), plotted all data points (one mark), drew two best-fit lines correctly (one mark), and demonstrated how to determine the temperature difference using the extrapolation method (one mark). The candidate also labelled the most anomalous point and correctly explained it, referencing the key concept of 'incomplete mixing of the mixture', earning one mark.

Example candidate response—grade C

T/°C



Calculate the maximum temperature change at 2.5 minutes.

$$39.67 - 24.38 = \cancel{15.33} \quad 15.2 \quad \Delta T = \dots\dots\dots \cancel{15.3} \quad 15.2 \quad ^\circ\text{C} \quad [4]$$

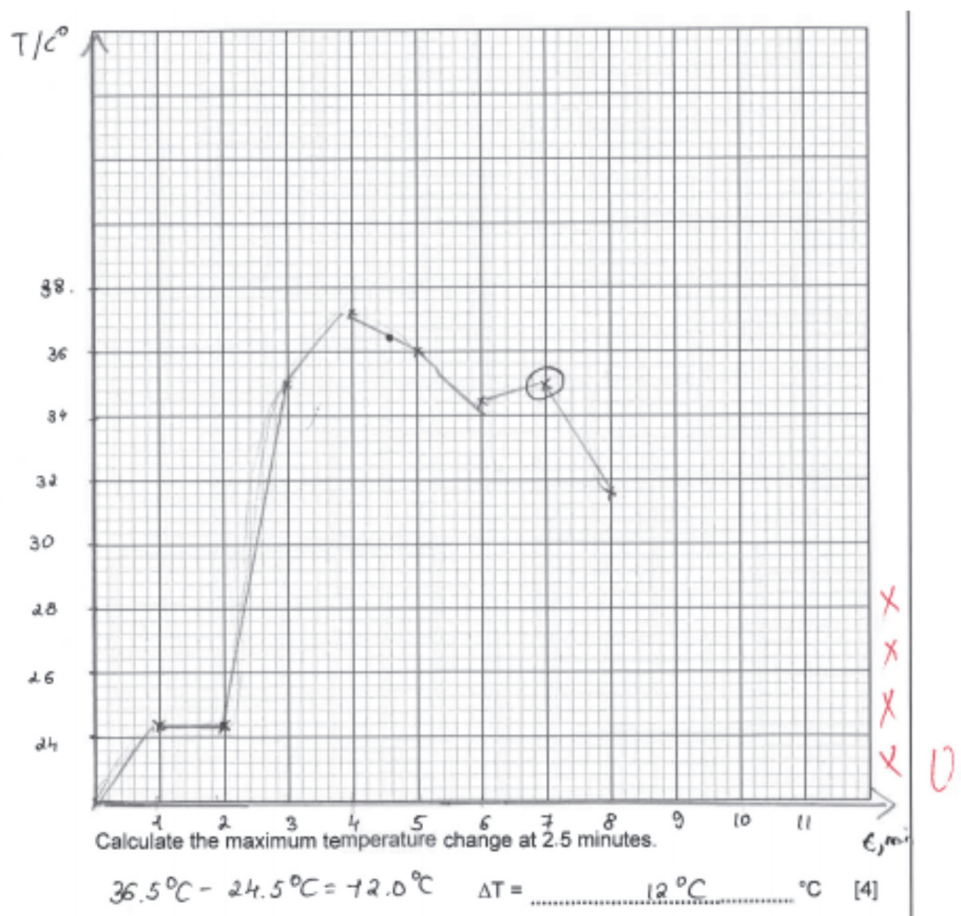
- (ii) Circle the most anomalous point on your graph and suggest a reason for the anomalous result.

Some unreacted CaO gave this burst of heat energy which might have happened because of [1] non-uniform stirring. ✓

Examiner's comment

The candidate earned four out of five marks. He labelled the X and Y axes with the correct units of measurement. However, he incorrectly set the scale for the Y axis, resulting in a loss of one mark. The candidate plotted all eight data points (earning one mark), drew two best-fit lines (one mark), and demonstrated how to determine the temperature difference using the extrapolation method (one mark). He also labelled the most anomalous point and correctly explained it, referencing the key concept of 'incomplete mixing of the mixture', earning one mark.

Example candidate response—grade E



- (ii) Circle the most anomalous point on your graph and suggest a reason for the anomalous result.

Maybe because of the fact that heat
has remained in the system. [1] X 0

Examiner's comment

The candidate did not earn any points for this question. He labelled the X and Y axes with the correct the units of measurement. However, he incorrectly set the scale for the Y axis. The candidate plotted all eight data points however the value at 0 minutes does not correspond to 24.0°C , and the value at 4 minutes does not correspond to 37.5°C . The candidate did not draw two straight best-fit lines. The candidate was unable to extrapolate the temperature difference at 2.5 minutes, which should lie within the interval 15.5 ± 2 . Although he highlighted the anomalous point, he could not explain his choice correctly.

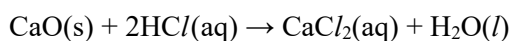
QUESTION 2(C)(III),(IV)

The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
Assume the reaction mixtures has a density of 1.0 g cm^3 .

Use the temperature change in (c)(i) to calculate the heat released in Experiment 2.

heat released = J [1]

Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_2 .



enthalpy change = kJ mol^{-1} [1]

Mark Scheme

2 (c) (iii)	Calcium carbonate results $q = \text{heat released} = 50 \times 4.2 \times \text{temperature increase (J)}$	[1]	Accept $0.02 \times 4.2 \times \text{temperature increase (J)}$ ecf from 2(c)(i)
2 (c) (iv)	enthalpy changes calculated = $(-q \div 0.025) / 1000$	[1]	answer must have negative sign not answer with +

Example candidate response—grade A

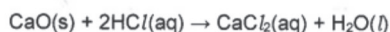
- (iii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$.
Assume the reaction mixtures has a density of 1.0 g cm^3 .
Use the temperature change in (c)(i) to calculate the heat released in Experiment 2.

$$\Delta t = 15^\circ\text{C}$$

$$Q = cm \Delta t = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1} \cdot 15^\circ\text{C} \cdot 50 \text{ g} = 3150 \text{ J}$$

heat released = 3150 J [1] ✓ 1

- (iv) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_2 .



$$n = \frac{1.4 \text{ g}}{(40 + 16) \text{ g mol}^{-1}} = 0.025$$

$$\Delta H = \frac{-3150 \text{ J}}{0.025 \text{ mol}} = -126000 \text{ J mol}^{-1}$$

enthalpy change = -126 kJ mol^{-1} [1] ✓ 1

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Use

Examiner's comment

The candidate earned two out of two marks for this question. He successfully completed all calculations in this question: the amount of heat released and the value of the enthalpy change for the reaction in kJ/mol.

Example candidate response—grade C

(iii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$. Assume the reaction mixture has a density of 1.0 g cm^3. Use the temperature change in (c)(i) to calculate the heat released in Experiment 2. $q = mc\Delta T$ $q = 1.4 \times 4.2 \times 15.18 = 89.26 \text{ J}$ heat released = <u>89.26</u> J [1] (iv) Calculate the enthalpy change, in kJ mol^{-1}, for the reaction below, ΔH_2. $\text{CaO(s)} + 2\text{HCl(aq)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{H}_2\text{O(l)}$ $\Delta H = - \frac{89.26}{0.025} = -3570.4 \text{ J/mol}$ $n = \frac{1.4}{40 + 16} = 0.025$ enthalpy change = <u>-3.57</u> kJ mol^{-1} [1]		For Examiner's Use 0 1
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Examiner's comment

The candidate earned one out of two marks. When calculating the amount of heat released during the reaction, he mistakenly used the mass of the dry substance instead of the mass of the solution, which led to an incorrect calculation, resulting in a loss of one mark. However, he was then able to calculate the enthalpy change, allowing for the application of ecf, earning one mark.

Example candidate response—grade E

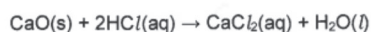
- (iii) The specific heat capacity of the reaction mixture, $c = 4.2 \text{ J } ^\circ\text{C}^{-1} \text{ g}^{-1}$. Assume the reaction mixture has a density of 1.0 g cm^3 .

Use the temperature change in (c)(i) to calculate the heat released in Experiment 2.

$$Q = mc\Delta T = 1,4 \cdot 4,2 \cdot 14 = 82,32$$

heat released = 82,3 J [1] X 0

- (iv) Calculate the enthalpy change, in kJ mol^{-1} , for the reaction below, ΔH_2 .



$$\Delta H = \frac{Q}{n} = \frac{-82,3 \text{ J}}{0,15 \text{ mol}} = \underline{548,66 \text{ J mol}^{-1}}$$

$n = \frac{50}{1000} \cdot 3 = 0,15$ enthalpy change = -0,54 kJ mol^{-1} [1] X 0

For
Examiner
Use

Examiner's comment

The candidate did not earn any marks for this question. When calculating the amount of heat released during the reaction, he mistakenly used the mass of the dry substance instead of the mass of the solution, resulting in an incorrect calculation and the loss of one mark. When determining the enthalpy change for the reaction, he determined the amount of substance in excess but correctly calculated the enthalpy change considering the sign for an exothermic reaction. No mark is awarded for such answer.

QUESTION 2(D),(E)

- (d) (i) Use the equations in parts (b)(iii) and (c)(iv) to construct the Hess' Law cycle to show how you could calculate the enthalpy change for the decomposition of calcium carbonate, ΔH reaction.

Label the arrows in your cycle.

[2]

- (ii) Use your values for enthalpy change from Experiment 1 and Experiment 2 to calculate the enthalpy change for the decomposition of calcium carbonate, ΔH_r

ΔH_r kJ mol⁻¹ [1]

- (e) The procedure in Experiment 2 is repeated using the same mass of calcium oxide.

This time 50 cm³ of 4.0 mol dm⁻³ HCl is used instead of 50 cm³ of 3.0 mol dm⁻³ HCl.

Compare the maximum temperature change in this case with the temperature change in (c)(i).

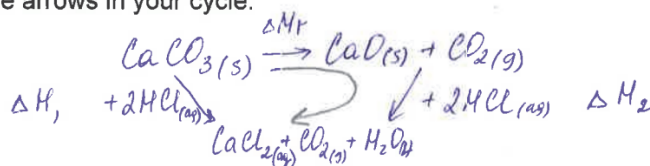
.....
 [1]

Mark Scheme

2 (d)(i)	Attempt at Hess cycle diagram All correct species downward arrow from CaCO₃, labelled with value from (b)(i) or ΔH_1 downward arrow from CaO, labelled with value from (b)(ii) or ΔH_2	1 1 [2]	ignore state symbols
2 (d)(ii)	ΔH_r value = ($\Delta H_{\text{experiment 1}} - \Delta H_{\text{experiment 2}}$)	[1] [1]	sign must be correct from the enthalpy changes, but ignore missing + if the answer comes out positive
2 (e)	temperature rise would be same	[1] [1]	

Example candidate response—grade A

- (d) (i) Use the equations in parts (b)(iii) and (c)(iv) to construct the Hess' Law cycle to show how you could calculate the enthalpy change for the decomposition of calcium carbonate, ΔH reaction. Label the arrows in your cycle.



$$\Delta H_1 + \Delta H_2 = \Delta H_3$$

$$\Delta H_r = \Delta H_1 - \Delta H_2$$

✓
✓

[2]

- (ii) Use your values for enthalpy change from Experiment 1 and Experiment 2 to calculate the enthalpy change for the decomposition of calcium carbonate, ΔH_r

$$\Delta H_r = \Delta H_1 - \Delta H_2 = (-4.9 - (-16.8)) \text{ kJ mol}^{-1} = 11.9 \text{ kJ mol}^{-1}$$

$$\Delta H_f = -4.9 - (-21.0) = 16.1 \text{ kJ mol}^{-1}$$

✓ conf.

$$\Delta H_f \quad \text{11.9} \quad 16.1 \quad \text{kJ mol}^{-1} \quad [1]$$

For
Examiner's
Use

- (e) The procedure in Experiment 2 is repeated using the same mass of calcium oxide.

This time 50 cm^3 of $4.0 \text{ mol dm}^{-3} \text{ HCl}$ is used instead of 50 cm^3 of $3.0 \text{ mol dm}^{-3} \text{ HCl}$.

Compare the maximum temperature change in this case with the temperature change in (c)(i).

temperature^{change} will not change, it will be approximately the same. [1]

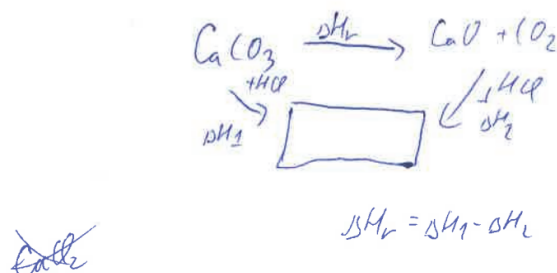
[1]

Examiner's comment

The candidate earned four out of four marks for this question. He received two marks for correctly constructing the Hess' Law cycle, ensuring that all arrows were appropriately labelled. He correctly calculated ΔH_r for the decomposition reaction of calcium carbonate using the formula derived from the Hess' Law cycle. He also correctly predicted how a change in acid concentration would affect the temperature change.

Example candidate response—grade C

- (d) (i) Use the equations in parts (b)(iii) and (c)(iv) to construct the Hess' Law cycle to show how you could calculate the enthalpy change for the decomposition of calcium carbonate, ΔH_r reaction. Label the arrows in your cycle.



XX
✓✓

[2]

For
Examiner's
Use

- (ii) Use your values for enthalpy change from Experiment 1 and Experiment 2 to calculate the enthalpy change for the decomposition of calcium carbonate, ΔH_r .

$$\Delta H_r = \Delta H_1 - \Delta H_2 = 4,9 - \frac{22,4}{2} = -16,8 \text{ kJ mol}^{-1} - 17,5 \text{ kJ mol}^{-1}$$

$$\Delta H_r = -46,8 - 17,5 \text{ kJ mol}^{-1} \quad [1]$$

✓
✓ect

- (e) The procedure in Experiment 2 is repeated using the same mass of calcium oxide.

This time 50 cm³ of 4.0 mol dm⁻³ HCl is used instead of 50 cm³ of 3.0 mol dm⁻³ HCl.

Compare the maximum temperature change in this case with the temperature change in (c)(i).

Due to being more concentrated maximum temperature change in this case
 case will be higher compared to HCl of concentration 3.0 mol dm⁻³

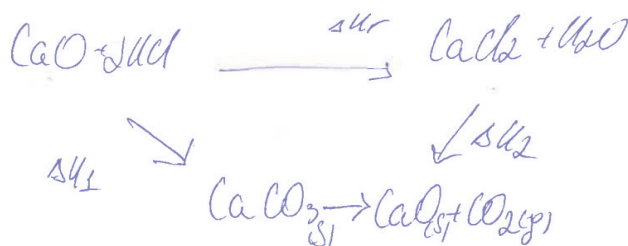
[1]

Examiner's comment

The candidate earned two out of four marks for this question. He did not complete the Hess' Law cycle, but correctly indicated the direction of the arrows and signed them, earning one mark. He correctly calculated ΔH_r for the decomposition reaction of calcium carbonate using the formula derived from the Hess' Law cycle. The candidate incorrectly predicted how a change in hydrochloric acid concentration would affect the temperature change.

Example candidate response—grade E

- (d) (i) Use the equations in parts (b)(iii) and (c)(iv) to construct the Hess' Law cycle to show how you could calculate the enthalpy change for the decomposition of calcium carbonate, ΔH reaction. Label the arrows in your cycle.



X
X
[2] 0

- (ii) Use your values for enthalpy change from Experiment 1 and Experiment 2 to calculate the enthalpy change for the decomposition of calcium carbonate, ΔH_r

$$-\Delta H_1 - \Delta H_2 = +28.4 - 139.4 = -105$$

$$\Delta H_r = -105 \text{ kJ mol}^{-1} \quad [1]$$

X
0

- (e) The procedure in Experiment 2 is repeated using the same mass of calcium oxide.

This time 50 cm³ of 4.0 mol dm⁻³ HCl is used instead of 50 cm³ of 3.0 mol dm⁻³ HCl.

Compare the maximum temperature change in this case with the temperature change in (c)(i).

maximum temperature change would be higher
due to increasing rate of all reactions, [1]

X
0

Examiner's comment

The candidate did not earn any marks for this question because he incorrectly constructed the Hess' Law cycle and, accordingly, failed to calculate the enthalpy change of the decomposition reaction of calcium carbonate. The candidate also incorrectly predicted how a change in hydrochloric acid concentration would affect the temperature change.